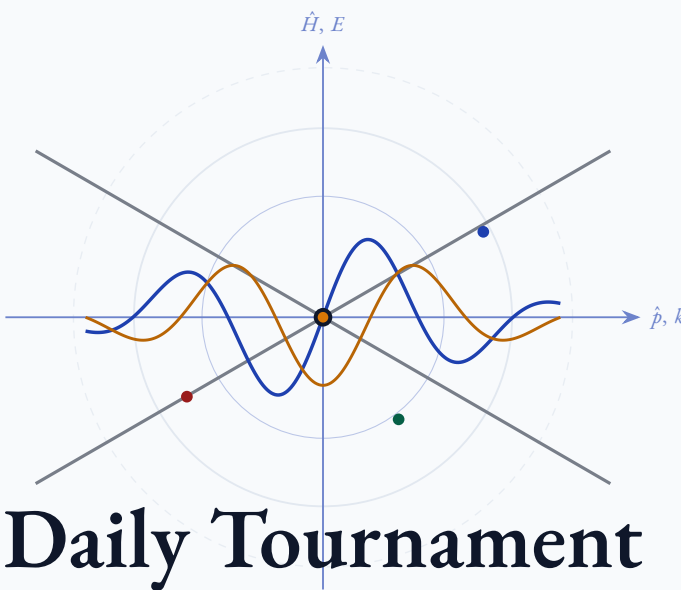




# Daily Tournament

GRAND ARENA OF THEORETICAL PHYSICS

*“Today’s contestants are ready: 604 Calculus  $\times$  811 Quantum Mechanics”*

Closed-book · Pen-and-paper derivation · No shortcuts

|                |                                                              |
|----------------|--------------------------------------------------------------|
| Date           | [YYYY-MM-DD]   Week [X] (Day [N])                            |
| Modules        | 604: [module]   811: [module]                                |
| Error Protocol | 5-Factor: Knowledge   Modelling   Logic   Calculation   Time |
| Target         | Standard 5.5–6.5 h   Minimum Defence 3.5 h                   |

## Theoretical Physics Training · Daily Tournament

[YYYY-MM-DD] | Week [X] (Day [N]) | Phase: [phase]

## §1 Tactical Briefing &amp; Warm-Up Retrieval

- 604: [今日高数任务、核心方法与目标题型]
- 811: [今日量力物理模型、算符代数与推导目标]
- Supporting: 英语: 真题精读 1 篇与长难句剖析; 政治: 30 分钟专项选择题/考点梳理。

**Momentum Defence Line:** Standard push 5.5–6.5 h | Minimum defence 3.5 h (must secure: 811 retrieval + M1 + English reading).

**Warm-Up Retrieval Checklist** (5–10 min): without notes, write on scratch paper:

☐ [核心公式 / 定理条件 1] ☐ [核心公式 / 定理条件 2] ☐ [核心公式 / 定理条件 3]

## Warm-Up Retrieval Workspace | 白纸破冰草稿区

5–10 分钟闭卷盲写: 在下方空白区域手推上述 3 个核心公式/边界条件, 快速越过启动能垒。

§2    Core Problem Set

Complete independently under timed conditions

Problem M1 | 604 Calculus: [problem type]

Suggested time: [25] min

此处填入 604 高等数学题目正文。写清展开阶数与适用条件、几何或代数转化、规范步骤, 并给出最终化简结果。

Problem Q1 | 811 Quantum Mechanics: [model]

Suggested time: [35] min

此处填入 811 量子力学题目正文。写清物理情景与边界条件、哈密顿量  $\hat{H}$  或算符表象, 求解  $\psi(x)$  与  $\hbar$  量纲自检。

Extension | Conceptual Analysis

Optional · Suggested time: 10 min

此处填入一道精简概念辨析, 或前序错题的微变式。独立作答合计主问题限时不超过 80 分钟。

§3    Model Solutions & Commentary

Complete your independent attempt on scratch paper before consulting this page.

■ M1 · Model Solution & Commentary

**Strategy:** 阐明解题突破口与数学转化思想,指出主导项或几何结构。

**Derivation:** 填入关键公式与严密推导过程,算到最终化简形式。

【关键推导公式】

■ **[Common Error]** 关键避坑提醒:例如等价无穷小误用于加减、洛必达未验证未定式。

■ Q1 · Model Solution & Commentary

**Physical Picture:** 说明波函数演化、对称性、守恒量或算符对易关系。

**Derivation:** 按「物理条件 → 哈密顿量/表象 → 边界或对称 → 计算 → 物理解释」书写完整推导。

【关键推导公式】

■ **[Checkpoint]** 自检:归一化系数、边界连续性、内积复共轭、时间演化相位符号、量纲。

§4 Daily Debrief & Reflection

Post-Match Review (fill in before sleep, 5 min)

|              |                                                                                                                                                                                                                                                                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 604 Calculus | Time spent: _____ min   M1 status: <input type="checkbox"/> Independent <input type="checkbox"/> Used hints <input type="checkbox"/> Incomplete<br>Error type: <input type="checkbox"/> Knowledge gap <input type="checkbox"/> Modelling <input type="checkbox"/> Logic <input type="checkbox"/> Calculation <input type="checkbox"/> Time                 |
| 811 QM       | Time spent: _____ min   Q1 status: <input type="checkbox"/> Independent <input type="checkbox"/> Used hints <input type="checkbox"/> Incomplete<br>Error type: <input type="checkbox"/> Knowledge gap <input type="checkbox"/> Physical model <input type="checkbox"/> Operator algebra <input type="checkbox"/> Calculation <input type="checkbox"/> Time |
| Planning     | Spaced retrieval in 3 days: _____<br>Today's assessment: <input type="checkbox"/> Peak performance <input type="checkbox"/> Steady progress <input type="checkbox"/> Minimum defence mode                                                                                                                                                                  |

Key Derivation Insights & Traps | 突破口与避坑沉淀

Core Breakthrough (今日核心突破口与关键一步):

\_\_\_\_\_

Fatal Trap & Common Error (致命陷阱与避坑警示):

\_\_\_\_\_

Day +3 Action Item (3 天后间隔重做提取的具体题号与断点):

\_\_\_\_\_